Thermodynamics and Statistical Physics

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Homework 13

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Show all essential steps in your derivations.

Angular average and the Stefan–Boltzmann law

In the main text we derived the spectral energy density of blackbody radiation,

$$u(\omega, T)\, d\omega = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{d\omega}{e^{\beta\hbar\omega} - 1}. \tag{1}$$

We also found the total energy density

$$u(T) = \int_0^\infty u(\omega, T)\, d\omega = \frac{\pi^2}{15} \frac{T^4}{\hbar^3 c^3}. \tag{2}$$

In this problem we derive the energy flux crossing a surface.
Consider an isotropic photon gas inside a cavity. Let dA be a small surface element, and let $\hat{\mathbf{n}}$ be its outward normal direction. Photons move with speed c. Let θ be the angle between the photon velocity and $\hat{\mathbf{n}}$.

(a) Explain why only photons with $0 \le \theta \le \pi/2$ contribute to the outward flux through the surface. For photons in the frequency interval $(\omega, \omega + d\omega)$, the energy density per solid angle is $u(\omega, T)d\omega/(4\pi)$. Show that the energy crossing the area dA during time dt, from directions within $d\Omega$, is

$$dE = u(\omega, T)\, d\omega \frac{d\Omega}{4\pi}\, dA\, c \cos\theta\, dt. \tag{3}$$

(b) Define $j(\omega, T)d\omega$ as the outward energy flux carried by photons with frequencies in $(\omega, \omega + d\omega)$. Integrate over the outward hemisphere and show that

$$j(\omega, T)\, d\omega = c\, u(\omega, T)\, d\omega \int_{\text{outward}} \frac{d\Omega}{4\pi} \cos\theta = \frac{c}{4} u(\omega, T)\, d\omega. \tag{4}$$

(c) Integrate over frequency to obtain

$$j(T) = \int_0^\infty j(\omega, T)\, d\omega = \frac{c}{4} u(T). \tag{5}$$

Then use Eq. (2) to derive the Stefan–Boltzmann law,

$$j(T) = \sigma T^4, \qquad \sigma = \frac{\pi^2}{60} \frac{1}{\hbar^3 c^2}. \tag{6}$$

Solution

(a) The outward normal to the surface dA is defined as the reference direction $\theta = 0$. For a photon to cross the surface from the inside to the outside, its velocity vector must have a positive component along the normal $\hat{\mathbf{n}}$. This means $v_n = c\cos\theta > 0$, which restricts the angle to $0 \le \theta \le \pi/2$. Photons with $\pi/2 < \theta \le \pi$ are moving away from the surface (inward) and do not contribute to the outward flux.

Consider the photons with frequencies in $(\omega, \omega + d\omega)$. Since the radiation field is isotropic, the energy density is uniformly distributed across all solid angles. The total solid angle is 4π, so the energy density per unit solid angle is $u(\omega, T)d\omega/(4\pi)$. The energy contained in a solid angle $d\Omega$ is therefore $u(\omega, T)d\omega\frac{d\Omega}{4\pi}$. In a time interval dt, the photons moving within $d\Omega$ at angle θ sweep out a slanted cylinder of length cdt and base area dA. The volume of this cylinder is $dV = dA(cdt\cos\theta) = c\cos\theta\, dA\, dt$. The energy crossing dA from this solid angle is the energy density in $d\Omega$ multiplied by this volume:

$$dE = \left[u(\omega, T)d\omega\frac{d\Omega}{4\pi}\right] \times (c\cos\theta\, dA\, dt) = u(\omega, T)\, d\omega\, \frac{d\Omega}{4\pi}\, dA\, c\cos\theta\, dt. \tag{7}$$

(b) The energy flux $j(\omega, T)d\omega$ is defined as the energy crossing per unit area per unit time, so $j(\omega, T)d\omega = dE/(dA\, dt)$. From part (a), integrating over all outward directions (the outward hemisphere):

$$j(\omega, T)\, d\omega = \int_{\text{hemisphere}} \frac{dE}{dA\, dt} = \int_{\theta=0}^{\pi/2}\int_{\phi=0}^{2\pi} u(\omega, T)\, d\omega\, \frac{\sin\theta\, d\theta\, d\phi}{4\pi}\, c\cos\theta. \tag{8}$$

Factoring out the constants:

$$j(\omega, T)\, d\omega = c\, u(\omega, T)\, d\omega \int_0^{2\pi} \frac{d\phi}{4\pi} \int_0^{\pi/2} \cos\theta\sin\theta\, d\theta. \tag{9}$$

The ϕ integral gives 2π. The θ integral gives $\int_0^{\pi/2} \frac{1}{2}\sin(2\theta)d\theta = \frac{1}{2}$. Thus:

$$\int_{\text{outward}} \frac{d\Omega}{4\pi}\cos\theta = \frac{2\pi}{4\pi} \times \frac{1}{2} = \frac{1}{4}. \tag{10}$$

Therefore, we obtain the spectral flux result:

$$j(\omega, T)\, d\omega = \frac{c}{4} u(\omega, T)\, d\omega. \tag{11}$$

(c) Integrating the spectral flux over all frequencies gives the total outward energy flux $j(T)$:

$$j(T) = \int_0^{\infty} j(\omega, T)\, d\omega = \int_0^{\infty} \frac{c}{4} u(\omega, T)\, d\omega = \frac{c}{4} u(T). \tag{12}$$

We are given the total energy density of blackbody radiation $u(T) = \frac{\pi^2}{15}\frac{T^4}{\hbar^3 c^3}$. Substituting this into the flux equation yields:

$$j(T) = \frac{c}{4}\left(\frac{\pi^2}{15}\frac{T^4}{\hbar^3 c^3}\right) = \frac{\pi^2}{60}\frac{1}{\hbar^3 c^2}T^4. \tag{13}$$

Defining the constant $\sigma = \frac{\pi^2}{60\hbar^3 c^2}$, we obtain the Stefan-Boltzmann law:

$$j(T) = \sigma T^4. \tag{14}$$

What does “black” mean in blackbody radiation?

A **blackbody** is defined as a body that absorbs all radiation incident on it. In this problem we show that, in thermal equilibrium, a perfect absorber is also a universal emitter.

Consider a small body placed inside a cavity whose radiation field is in thermal equilibrium at temperature T. Let $j(\omega, T)\, d\omega$ be the incident energy per unit area per unit time in the frequency interval $(\omega, \omega + d\omega)$. According to the previous problem,

$$j(\omega, T) = \frac{c}{4} u(\omega, T), \tag{15}$$

where $u(\omega, T)$ is the Planck spectral energy density. Hence $j(\omega, T)$ is universal: it is independent of the material inserted into the cavity.

Let $\alpha(\omega, T)$ be the absorptivity of the body, namely the fraction of incident radiation at frequency ω that is absorbed. Let $e(\omega, T)\, d\omega$ be the energy emitted per unit area per unit time in the same frequency interval.

(a) Explain why energy balance must hold separately in each frequency interval in thermal equilibrium. In other words, explain why the body cannot have a net gain of energy in one frequency interval and a net loss in another frequency interval.

Hint: If such a compensation between different frequencies were allowed, one could use frequency filters to separate the two frequency ranges and produce a net energy flow between systems at the same temperature. This would violate the second law.

(b) Write the absorbed energy per unit area per unit time in the interval $(\omega, \omega + d\omega)$. Using the frequency-by-frequency energy balance from part (a), show that

$$e(\omega, T) = \alpha(\omega, T)\, j(\omega, T). \tag{16}$$

(c) Now compare two different bodies placed in the same cavity at the same temperature. Show that

$$\frac{e_A(\omega, T)}{\alpha_A(\omega, T)} = \frac{e_B(\omega, T)}{\alpha_B(\omega, T)}. \tag{17}$$

Thus e/α is independent of the material. This is **Kirchhoff’s law of thermal radiation**.

(d) A perfect absorber has $\alpha(\omega, T) = 1$. Explain why such a body emits the universal thermal spectrum. Here universal means independent of material. Why does this justify the name **blackbody radiation**, even though a hot blackbody emits light?

Solution

(a) In thermal equilibrium, the principle of **detailed balance** demands that net energy flow must be zero not just overall, but for every specific microscopic process. If a body had a net gain of energy in one frequency interval ω_1 and a net loss in another interval ω_2 (while keeping the total energy constant), we could place this body inside a cavity and surround it with a perfectly reflective filter that strictly transmits ω_1 but reflects ω_2. The body would then continuously absorb more energy than it emits at ω_1, draining energy from the cavity radiation field at ω_1. Meanwhile, it would emit more than it absorbs at ω_2, pumping energy into the cavity field at ω_2. This would spontaneously drive the cavity radiation out of equilibrium (creating temperature gradients across frequencies or spatial regions) without any external work, directly violating the Second Law of Thermodynamics. Therefore, energy balance must hold exactly within every infinitesimal frequency interval.

(b) The total incident energy per unit area per unit time in the interval $(\omega, \omega + d\omega)$ is $j(\omega, T)d\omega$. By definition of the absorptivity $\alpha(\omega, T)$, the fraction of this incident energy that is absorbed by the body is $\alpha(\omega, T)$. Therefore, the absorbed energy per unit area per unit time is:

$$\text{Absorbed} = \alpha(\omega, T)\, j(\omega, T)\, d\omega. \tag{18}$$

The emitted energy per unit area per unit time in the same interval is $e(\omega, T)\, d\omega$. According to the frequency-by-frequency energy balance established in part (a), the absorbed energy must exactly equal the emitted energy to maintain thermal equilibrium:

$$e(\omega, T)\, d\omega = \alpha(\omega, T)\, j(\omega, T)\, d\omega \implies e(\omega, T) = \alpha(\omega, T)\, j(\omega, T). \tag{19}$$

(c) Consider two different bodies, A and B, placed in the same thermal equilibrium cavity at temperature T. Since the incident radiation flux $j(\omega, T) = \frac{c}{4} u(\omega, T)$ depends strictly on the cavity temperature T and is completely independent of any material properties, both bodies experience the exact same $j(\omega, T)$. Applying the equilibrium condition from part (b) to each body:

$$e_A(\omega, T) = \alpha_A(\omega, T)\, j(\omega, T) \implies j(\omega, T) = \frac{e_A(\omega, T)}{\alpha_A(\omega, T)} \tag{20}$$

$$e_B(\omega, T) = \alpha_B(\omega, T)\, j(\omega, T) \implies j(\omega, T) = \frac{e_B(\omega, T)}{\alpha_B(\omega, T)} \tag{21}$$

Equating the two expressions for the universal incident flux $j(\omega, T)$ gives Kirchhoff's law:

$$\frac{e_A(\omega, T)}{\alpha_A(\omega, T)} = \frac{e_B(\omega, T)}{\alpha_B(\omega, T)} = j(\omega, T). \tag{22}$$

(d) A perfect absorber (a "blackbody") absorbs all incident radiation, meaning $\alpha(\omega, T) = 1$ for all frequencies. Substituting $\alpha = 1$ into Kirchhoff's law gives:

$$e_{\text{blackbody}}(\omega, T) = 1 \times j(\omega, T) = \frac{c}{4} u(\omega, T). \tag{23}$$

Because $j(\omega, T)$ is determined solely by the universal Planck spectrum of the cavity $u(\omega, T)$, the emitted radiation $e_{\text{blackbody}}(\omega, T)$ is also exactly this universal thermal spectrum. It depends only on the temperature T, not on the material composition or structure of the perfect absorber. It is called "blackbody radiation" because an ideal object that looks perfectly black at room temperature (absorbing all visible light) is the exact same theoretical object that will perfectly emit this universal thermal spectrum when heated. The name refers to its ideal property as a perfect absorber (black), which inevitably makes it the perfect universal emitter.

Adiabatic expansion of blackbody radiation

In the main text we considered an adiabatic expansion of a cavity in which all linear dimensions are multiplied by a factor α. We argued microscopically that the photon frequencies scale as $\omega \to \omega/\alpha$, so a blackbody radiation field remains a blackbody, but with a lower temperature $T \to T/\alpha$.

In this problem we rederive the same result from thermodynamics. To make contact with cosmology, we use $R(t)$ to denote the scale factor of the universe. It describes how large-scale distances change with time. If $R(t)$ changes by a factor α, then a typical length changes by the same factor, and a typical volume changes by α^3. Thus the volume scales as

$$V(t) \propto R(t)^3. \tag{24}$$

Equivalently, if t_0 is the present time and $t < t_0$ is an earlier time, then

$$\frac{V(t)}{V(t_0)} = \left(\frac{R(t)}{R(t_0)}\right)^3. \tag{25}$$

Assume that the radiation remains in thermal equilibrium and expands adiabatically. From the theory of blackbody radiation, we have

$$U(t) = a\,V(t)\,T(t)^4, \qquad P(t) = \frac{1}{3}aT(t)^4, \qquad a = \frac{\pi^2}{15}\frac{1}{\hbar^3 c^3}. \tag{26}$$

(a) For an adiabatic expansion, the first law gives

$$dU = -P\,dV. \tag{27}$$

Use Eq. (26) to show that

$$V(t)T(t)^3 = \text{constant}. \tag{28}$$

(b) Using $V(t) \propto R(t)^3$, show that the temperature of the blackbody radiation scales as

$$T(t) \propto R(t)^{-1}. \tag{29}$$

(c) Show that the total radiation energy inside the expanding volume scales as

$$U(t) \propto R(t)^{-1}. \tag{30}$$

Explain the meaning of Eq. (27) in this setting. In particular, why should the decrease of U not be interpreted as energy being carried away to something outside the universe?

Solution

(a) The first law of thermodynamics for an adiabatic process ($dQ = 0$) is $dU = -P\,dV$. For blackbody radiation, we are given $U = aVT^4$ and $P = \frac{1}{3}aT^4$. Taking the total differential of $U(T, V)$:

$$dU = d(aVT^4) = aT^4\,dV + 4aVT^3\,dT. \tag{31}$$

Substitute dU and P into the first law:

$$aT^4\,dV + 4aVT^3\,dT = -\left(\frac{1}{3}aT^4\right)dV. \tag{32}$$

Dividing both sides by $aT^3 \neq 0$:

$$T\,dV + 4V\,dT = -\frac{1}{3}T\,dV \implies \frac{4}{3}T\,dV + 4V\,dT = 0. \tag{33}$$

Dividing by $4TV$ gives:

$$\frac{1}{3}\frac{dV}{V} + \frac{dT}{T} = 0. \tag{34}$$

Integrating both sides yields:

$$\frac{1}{3}\ln V + \ln T = \text{constant} \implies \ln(V^{1/3}T) = \text{constant} \implies VT^3 = \text{constant}. \tag{35}$$

(b) From part (a), we have $T(t)^3 \propto V(t)^{-1}$. Taking the cube root gives $T(t) \propto V(t)^{-1/3}$. We are given the relation between volume and the scale factor: $V(t) \propto R(t)^3$. Substituting this into the temperature relation:

$$T(t) \propto \left[R(t)^3\right]^{-1/3} = R(t)^{-1}. \tag{36}$$

Thus, the temperature of the radiation field falls exactly inversely with the expansion scale factor.

(c) The total radiation energy is $U(t) = aV(t)T(t)^4$. Substitute the scaling behaviors $V(t) \propto R(t)^3$ and $T(t) \propto R(t)^{-1}$:

$$U(t) \propto R(t)^3\left[R(t)^{-1}\right]^4 = R(t)^3R(t)^{-4} = R(t)^{-1}. \tag{37}$$

The total energy $U(t)$ decreases as the universe expands. The meaning of $dU = -P\,dV$ in this cosmological setting is that the expanding radiation gas does continuous $P\,dV$ work on the boundaries of the comoving volume element it occupies. As the universe expands, gravity acts as the "piston"; the radiation loses internal energy by doing work against the gravitational expansion of the fabric of spacetime itself. The decrease of U should not be interpreted as energy leaking "outside the universe" because there is no "outside." The energy simply changes form, being expended to perform mechanical work that drives the cosmological expansion.

Number fluctuations in a degenerate Bose gas

Consider a spinless ideal Bose gas in a three-dimensional box of volume $V = L^3$, with periodic boundary conditions. We focus on the condensed regime $T < T_c$. In the first part of the problem, treat the excited states $\mathbf{k} \neq 0$ as a grand-canonical subsystem with chemical potential $\mu = 0$. The occupation number of an excited state is then

$$\langle n_{\mathbf{k}} \rangle = \frac{1}{e^{\beta \epsilon_{\mathbf{k}}} - 1}, \qquad \epsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}, \qquad \mathbf{k} \neq 0. \tag{38}$$

(a) For a single bosonic one-particle state α, use the grand canonical distribution to show that

$$\langle (\Delta n_\alpha)^2 \rangle = \langle n_\alpha \rangle \left(1 + \langle n_\alpha \rangle\right). \tag{39}$$

(b) Define the number of particles in excited states by

$$N_{\text{exc}} = \sum_{\mathbf{k} \neq 0} n_{\mathbf{k}}. \tag{40}$$

Using the independence of different one-particle states in the grand canonical ensemble, show that

$$\langle (\Delta N_{\text{exc}})^2 \rangle = \sum_{\mathbf{k} \neq 0} \langle n_{\mathbf{k}} \rangle \left(1 + \langle n_{\mathbf{k}} \rangle\right). \tag{41}$$

(c) The first term in Eq. (41) is

$$\sum_{\mathbf{k} \neq 0} \langle n_{\mathbf{k}} \rangle = \langle N_{\text{exc}} \rangle. \tag{42}$$

Show that this term is proportional to V, and hence is of order N at fixed density.

Hint: Replace the sum by an integral,

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3} \int d^3k. \tag{43}$$

For $T < T_c$, the integral over k is finite; the only explicit dependence on the box size is the prefactor V.

(d) Now estimate the second term in Eq. (41). This term is more singular at small k. First expand the Bose–Einstein occupation number for small k. Since

$$\epsilon_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} \tag{44}$$

and $\beta \epsilon_{\mathbf{k}} \ll 1$, we have

$$\langle n_{\mathbf{k}} \rangle = \frac{1}{e^{\beta \epsilon_{\mathbf{k}}} - 1} \simeq \frac{T}{\epsilon_{\mathbf{k}}} = \frac{2mT}{\hbar^2 k^2}. \tag{45}$$

Thus $\langle n_{\mathbf{k}}\rangle^2$ behaves as $1/k^4$ at small k.

If we naively replace the sum by an integral down to $k = 0$, the integral is infrared divergent. In a finite box, however, wave vectors smaller than the smallest nonzero wave vector do not exist. We therefore introduce the infrared cutoff

$$k_{\min} = \frac{2\pi}{L}. \tag{46}$$

Using

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3}\int d^3k = \frac{V}{2\pi^2}\int_0^\infty k^2\,dk, \tag{47}$$

show that the leading small-k contribution is

$$\sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle^2 \simeq \frac{V}{2\pi^2}\left(\frac{2mT}{\hbar^2}\right)^2\int_{2\pi/L}^{k_*}\frac{dk}{k^2} = \frac{VL}{4\pi^3}\left(\frac{2mT}{\hbar^2}\right)^2 + O(V),$$

where k_* is a fixed upper cutoff for the small-k approximation, whose precise value does not affect the dependence on L which we care. Thus the anomalous contribution is proportional to

$$VL = L^4. \tag{48}$$

Finally, using fixed density $n = N/V$, show that

$$L^4 = \left(\frac{N}{n}\right)^{4/3}. \tag{49}$$

Conclude that, at fixed n and T,

$$\langle(\Delta N_{\rm exc})^2\rangle \sim N^{4/3}. \tag{50}$$

(e) For a finite homogeneous system coupled to a particle source, the fluctuation-response relation may be written as

$$\kappa_T = \frac{\langle(\Delta N)^2\rangle}{NnT}, \tag{51}$$

where $n = N/V$. Below T_c, the chemical potential is pinned at $\mu = 0$, and the singular number fluctuation comes from the excited cloud. Thus, for the leading finite-size singularity, we may replace $\langle(\Delta N)^2\rangle$ by $\langle(\Delta N_{\rm exc})^2\rangle$. Using Eq. (50), show that, at fixed density and temperature,

$$\kappa_T \sim N^{1/3}. \tag{52}$$

Explain why this is the finite-size version of the divergent isothermal compressibility of the ideal Bose gas below T_c.

(f) Now suppose that the total particle number is fixed:

$$N = N_0 + N_{\text{exc}}. \tag{53}$$

At low temperature $N_0 = O(N)$, so the condensate can act as a particle reservoir for the excited states. Explain why the grand-canonical treatment of the excited states remains valid for estimating the fluctuations of N_{exc}. Then show that

$$\langle(\Delta N_0)^2\rangle = \langle(\Delta N_{\text{exc}})^2\rangle \sim N^{4/3}, \tag{54}$$

and hence

$$\frac{\Delta N_0}{N_0} \sim N^{-1/3} \longrightarrow 0 \qquad (N \to \infty). \tag{55}$$

(g) Explain why it would be wrong to treat the condensate mode itself as an unconstrained grand-canonical mode. What fluctuation would one incorrectly obtain from

$$\langle(\Delta N_0)^2\rangle = \langle N_0\rangle\,(1 + \langle N_0\rangle)? \tag{56}$$

Solution

(a) In the grand canonical ensemble, the partition function for a single bosonic state α is $\Xi_\alpha = \sum_{n=0}^{\infty} e^{-\beta n(\epsilon_\alpha - \mu)}$. The expected particle number is:

$$\langle n_\alpha\rangle = \frac{1}{\beta}\frac{\partial \ln \Xi_\alpha}{\partial \mu} = \frac{1}{e^{\beta(\epsilon_\alpha-\mu)} - 1}. \tag{57}$$

Solving for the exponential gives $e^{\beta(\epsilon_\alpha-\mu)} = \frac{1}{\langle n_\alpha\rangle} + 1 = \frac{1+\langle n_\alpha\rangle}{\langle n_\alpha\rangle}$. The variance in particle number is obtained by taking a second derivative with respect to μ:

$$\langle(\Delta n_\alpha)^2\rangle = \frac{1}{\beta^2}\frac{\partial^2 \ln \Xi_\alpha}{\partial \mu^2} = \frac{1}{\beta}\frac{\partial\langle n_\alpha\rangle}{\partial \mu}. \tag{58}$$

Differentiating the Bose-Einstein distribution with respect to μ:

$$\frac{\partial\langle n_\alpha\rangle}{\partial \mu} = -\frac{1}{(e^{\beta(\epsilon_\alpha-\mu)} - 1)^2}\left(-\beta e^{\beta(\epsilon_\alpha-\mu)}\right) = \beta\langle n_\alpha\rangle^2 e^{\beta(\epsilon_\alpha-\mu)}. \tag{59}$$

Substituting the expression for the exponential derived earlier:

$$\frac{\partial\langle n_\alpha\rangle}{\partial \mu} = \beta\langle n_\alpha\rangle^2\left(\frac{1+\langle n_\alpha\rangle}{\langle n_\alpha\rangle}\right) = \beta\langle n_\alpha\rangle(1+\langle n_\alpha\rangle). \tag{60}$$

Multiplying by $1/\beta$, we obtain the required variance:

$$\langle(\Delta n_\alpha)^2\rangle = \langle n_\alpha\rangle(1+\langle n_\alpha\rangle). \tag{61}$$

(b) The total number of particles in excited states is $N_{\text{exc}} = \sum_{\mathbf{k}\neq 0} n_{\mathbf{k}}$. Because non-interacting bosons in different momentum states are statistically independent in the grand canonical ensemble, the variance of the sum is exactly the sum of the individual variances:

$$\langle(\Delta N_{\text{exc}})^2\rangle = \sum_{\mathbf{k}\neq 0}\langle(\Delta n_{\mathbf{k}})^2\rangle. \tag{62}$$

Substituting the single-state variance formula from part (a) (setting $\mu = 0$ for the condensed regime):

$$\langle(\Delta N_{\text{exc}})^2\rangle = \sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle(1 + \langle n_{\mathbf{k}}\rangle). \tag{63}$$

(c) The first term in the variance sum is simply the expectation value of the total number of excited particles:

$$\sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle = \langle N_{\text{exc}}\rangle. \tag{64}$$

We can estimate this term by converting the sum to an integral in the thermodynamic limit:

$$\langle N_{\text{exc}}\rangle = \sum_{\mathbf{k}\neq 0}\frac{1}{e^{\beta\hbar^2k^2/2m} - 1} \longrightarrow \frac{V}{(2\pi)^3}\int d^3k\frac{1}{e^{\beta\hbar^2k^2/2m} - 1}. \tag{65}$$

Below T_c, the integral $\int d^3k\ldots$ evaluates to a finite, size-independent constant that solely depends on temperature (it defines the critical density $n_c(T)$). Because the integral is finite and independent of V, we have $\langle N_{\text{exc}}\rangle \propto V$. At fixed density $n = N/V$, $V \propto N$. Therefore, this first term $\langle N_{\text{exc}}\rangle$ is of order $O(N)$.

(d) The second term in the variance sum is $\sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle^2$. For small k, using $\beta\epsilon_{\mathbf{k}} \ll 1$, we expand the exponential to get $\langle n_{\mathbf{k}}\rangle \simeq \frac{T}{\epsilon_{\mathbf{k}}} = \frac{2mT}{\hbar^2k^2}$. Replacing the sum with an integral and introducing the infrared box cutoff $k_{\text{min}} = 2\pi/L$:

$$\begin{aligned}
\sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle^2 &\simeq \frac{V}{2\pi^2}\int_{2\pi/L}^{k_*}k^2dk\left(\frac{2mT}{\hbar^2k^2}\right)^2 \\
&= \frac{V}{2\pi^2}\left(\frac{2mT}{\hbar^2}\right)^2\int_{2\pi/L}^{k_*}\frac{dk}{k^2} \\
&= \frac{V}{2\pi^2}\left(\frac{2mT}{\hbar^2}\right)^2\left[-\frac{1}{k}\right]_{2\pi/L}^{k_*} = \frac{V}{2\pi^2}\left(\frac{2mT}{\hbar^2}\right)^2\left(\frac{L}{2\pi} - \frac{1}{k_*}\right).
\end{aligned} \tag{66}$$

As $L \to \infty$, the upper cutoff $1/k_*$ is negligible compared to $L/2\pi$. Keeping only the leading divergence:

$$\sum_{\mathbf{k}\neq 0}\langle n_{\mathbf{k}}\rangle^2 \simeq \frac{VL}{4\pi^3}\left(\frac{2mT}{\hbar^2}\right)^2 + O(V). \tag{67}$$

This leading anomalous contribution is strictly proportional to VL. Since $V = L^3$, we have $VL = L^4$. At a fixed density $n = N/V$, the volume scales as $V = N/n$.

Therefore, the box length is $L = V^{1/3} = (N/n)^{1/3}$. Thus, the L^4 scaling translates to:

$$L^4 = \left(\left(\frac{N}{n}\right)^{1/3}\right)^4 = \left(\frac{N}{n}\right)^{4/3} \propto N^{4/3}. \tag{68}$$

Comparing the two terms: the first term is $O(N)$, while the second is $O(N^{4/3})$. In the thermodynamic limit $N \to \infty$, the anomalous $N^{4/3}$ term dominates the fluctuations. Thus:

$$\langle (\Delta N_{\text{exc}})^2 \rangle \sim N^{4/3}. \tag{69}$$

(e) Using the fluctuation-response relation $\kappa_T = \frac{\langle (\Delta N)^2 \rangle}{NnT}$ and substituting the dominant finite-size fluctuation of the excited states $\langle (\Delta N)^2 \rangle \approx \langle (\Delta N_{\text{exc}})^2 \rangle \sim N^{4/3}$, we obtain:

$$\kappa_T \sim \frac{N^{4/3}}{N \cdot n \cdot T} \sim \frac{N^{1/3}}{nT} \propto N^{1/3}. \tag{70}$$

As $N \to \infty$ (thermodynamic limit), the compressibility $\kappa_T \sim N^{1/3} \to \infty$. In the thermodynamic limit below T_c, the ideal Bose gas has a strictly infinite isothermal compressibility because the condensed phase acts as a bottomless reservoir: you can add or remove particles (changing the volume at fixed N, or N at fixed V) without changing the pressure at all (since $P \propto T^{5/2}$ is independent of density below T_c). For a finite box of size $L \propto N^{1/3}$, the addition of particles creates an infrared cutoff effect that slightly changes the pressure. The $N^{1/3}$ scaling of κ_T is the precise finite-size scaling manifestation of this macroscopic divergence; the system is highly compressible, but the true divergence is naturally rounded off by the finite box dimensions.

(f) When the total particle number N is strictly fixed, we have $N_0 + N_{\text{exc}} = N$, which means any fluctuation in N_{exc} is exactly balanced by an opposite fluctuation in N_0: $\Delta N_0 = -\Delta N_{\text{exc}}$. Below T_c, the condensate is macroscopic, meaning $\langle N_0 \rangle \sim O(N)$. Because it contains a vast number of particles, it behaves perfectly as an effective particle bath for the excited states (N_{exc}), exchanging particles without drastically affecting its own thermodynamic state. Therefore, the grand-canonical treatment with $\mu = 0$ remains highly accurate for evaluating the thermal fluctuations of the much smaller subsystem N_{exc}. Consequently, the variance of the condensate identically equals the variance of the excited states:

$$\langle (\Delta N_0)^2 \rangle = \langle (-\Delta N_{\text{exc}})^2 \rangle = \langle (\Delta N_{\text{exc}})^2 \rangle \sim N^{4/3}. \tag{71}$$

The relative fluctuation of the condensate is:

$$\frac{\Delta N_0}{\langle N_0 \rangle} \sim \frac{\sqrt{N^{4/3}}}{N} = \frac{N^{2/3}}{N} = N^{-1/3}. \tag{72}$$

As $N \to \infty$, $N^{-1/3} \to 0$. This ensures that the condensate fraction N_0/N remains a sharply defined, non-fluctuating thermodynamic macrovariable in the thermodynamic limit, confirming the validity of the phase transition.

(g) It would be fundamentally wrong to treat the condensate mode ($k = 0$) as an unconstrained grand-canonical mode because in the grand canonical ensemble below T_c, the chemical potential is rigidly pinned to $\mu = 0$. If we applied the single-state variance formula directly to the $\epsilon = 0$ state with $\mu = 0$, the mean occupation would diverge to infinity ($\langle N_0 \rangle = \frac{1}{1-1} \to \infty$). Even if we assumed $\langle N_0 \rangle$ took its macroscopic value $O(N)$ through a limiting procedure, the formula would yield:

$$\langle (\Delta N_0)^2 \rangle = \langle N_0 \rangle (1 + \langle N_0 \rangle) \approx \langle N_0 \rangle^2 \sim O(N^2). \tag{73}$$

This would imply a relative fluctuation $\Delta N_0 / \langle N_0 \rangle \sim O(1)$, meaning the number of particles in the condensate would wildly fluctuate by a macroscopic fraction of the entire system itself! This is physically absurd for an isolated system or one in thermal contact with a fixed environment, and it highlights the ”grand canonical catastrophe” for Bose-Einstein condensates: the non-interacting macroscopic ground state cannot act as its own particle reservoir.